Sales and Customer Analytics

Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.

1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           5
```

1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      19085
```

1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
                3817
```

1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
         1898
```

1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         5895
```

2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total_Amount column. All five functions can be combined into a single summary query.

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Order_Value,
  5      MIN(Total_Amount) AS Minimum_Order,
  6      MAX(Total_Amount) AS Maximum_Order
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------- ------------- -------------
           5       19085                3817          1898          5895
```

The output above is based on the sample data given at the end of this document (Myntra Fashion E-commerce Management System).

2.1 Customer-wise Sales Report

To find how much each customer has purchased:

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.First_Name,
 11      c.Last_Name
 12  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME TOTAL_PURCHASE
----------- ------------ --------------
       1001 Anitha Kumar            7793
       1003 Priya Nair              5895
       1002 Rahul Sharma            3499
       1004 Karthik Raja            1898

4 rows selected.
```

Purpose

This report shows:

- Customer ID
- Customer name
- Total purchase amount
- Customers arranged according to purchase amount

3. Find Top Customers Based on Purchase Amount

The SUM() function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts. FETCH FIRST 3 ROWS ONLY keeps just the top three customers.

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY
  9      c.Customer_ID,
 10      c.First_Name,
 11      c.Last_Name
 12  ORDER BY Total_Purchase DESC
 13  FETCH FIRST 3 ROWS ONLY;

CUSTOMER_ID CUSTOMER_NAME TOTAL_PURCHASE
----------- ------------ --------------
       1001 Anitha Kumar            7793
       1003 Priya Nair              5895
       1002 Rahul Sharma            3499

3 rows selected.
```

The customer appearing at the top has the highest purchase amount in the generated report.

4. Identify Best-Selling Products

The Order_Item table contains the quantity of each product ordered. Therefore, we can use SUM(Quantity) to find the total quantity sold for each product.

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold
  5  FROM Product p
  6  JOIN Order_Item oi
  7      ON p.Product_ID = oi.Product_ID
  8  GROUP BY
  9      p.Product_ID,
 10      p.Product_Name
 11  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME          TOTAL_QUANTITY_SOLD
---------- --------------------- -------------------
       102 HRX Printed T-Shirt                     5
       101 Roadster Cotton Shirt                   4
       103 Libas Anarkali Kurta                    3
       105 Puma Sports Cap                         2
       104 Puma Running Shoes                      1

5 rows selected.
```

Purpose

This query:

1. Connects Product and Order_Item.
2. Groups records according to product.
3. Calculates total quantity sold.
4. Displays products from highest to lowest quantity sold.

4.1 Product Sales Amount

Instead of only calculating quantity, we can calculate the total sales generated by each product. Since Order_Item stores Quantity and Unit_Price, the amount is Quantity multiplied by Unit_Price.

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold,
  5      SUM(oi.Quantity * oi.Unit_Price) AS Total_Sales
  6  FROM Product p
  7  JOIN Order_Item oi
  8      ON p.Product_ID = oi.Product_ID
  9  GROUP BY
 10      p.Product_ID,
 11      p.Product_Name
 12  ORDER BY Total_Sales DESC;

PRODUCT_ID PRODUCT_NAME          TOTAL_QUANTITY_SOLD TOTAL_SALES
---------- --------------------- ------------------- -----------
       103 Libas Anarkali Kurta                    3        5697
       101 Roadster Cotton Shirt                   4        5196
       104 Puma Running Shoes                      1        3499
       102 HRX Printed T-Shirt                     5        3495
       105 Puma Sports Cap                         2        1198

5 rows selected.
```

This gives both **quantity sold** and **sales amount**.

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

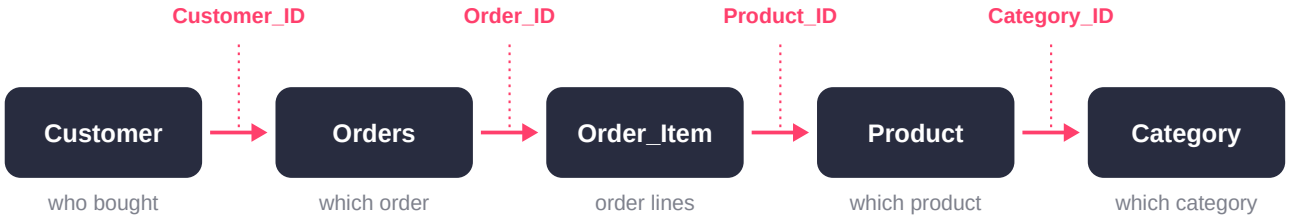
```
SQL> SELECT
  2      c.Category_ID,
  3      c.Category_Name,
  4      SUM(oi.Quantity) AS Total_Quantity_Sold,
  5      SUM(oi.Quantity * oi.Unit_Price) AS Total_Sales
  6  FROM Category c
  7  JOIN Product p
  8      ON c.Category_ID = p.Category_ID
  9  JOIN Order_Item oi
 10      ON p.Product_ID = oi.Product_ID
 11  GROUP BY
 12      c.Category_ID,
 13      c.Category_Name
 14  ORDER BY Total_Sales DESC;

CATEGORY_ID CATEGORY_NAME     TOTAL_QUANTITY_SOLD TOTAL_SALES
----------- ----------------- ------------------- -----------
          1 Men Topwear                         9        8691
          2 Women Ethnic Wear                   3        5697
          3 Footwear                            1        3499
          4 Accessories                         2        1198

4 rows selected.
```

6. Combined Business Sales Report

A complete sales report can combine customer, order, product, and category information. The report joins five tables in the order shown below.




```
SQL> SELECT
  2      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3      o.Order_ID,
  4      p.Product_Name,
  5      cat.Category_Name,
  6      oi.Quantity,
  7      oi.Quantity * oi.Unit_Price AS Subtotal
  8  FROM Customer c
  9  JOIN Orders o
 10      ON c.Customer_ID = o.Customer_ID
 11  JOIN Order_Item oi
 12      ON o.Order_ID = oi.Order_ID
 13  JOIN Product p
 14      ON oi.Product_ID = p.Product_ID
 15  JOIN Category cat
 16      ON p.Category_ID = cat.Category_ID
 17  ORDER BY o.Order_ID, p.Product_ID;

CUSTOMER_NAME ORDER_ID PRODUCT_NAME          CATEGORY_NAME     QUANTITY SUBTOTAL
------------- -------- --------------------- ----------------- -------- --------
Anitha Kumar      5001 Roadster Cotton Shirt Men Topwear               3     3897
Anitha Kumar      5001 Puma Sports Cap       Accessories               1      599
Rahul Sharma      5002 Puma Running Shoes    Footwear                  1     3499
Priya Nair        5003 HRX Printed T-Shirt   Men Topwear               3     2097
Priya Nair        5003 Libas Anarkali Kurta  Women Ethnic Wear         2     3798
Anitha Kumar      5004 HRX Printed T-Shirt   Men Topwear               2     1398
Anitha Kumar      5004 Libas Anarkali Kurta  Women Ethnic Wear         1     1899
Karthik Raja      5005 Roadster Cotton Shirt Men Topwear               1     1299
Karthik Raja      5005 Puma Sports Cap       Accessories               1      599

9 rows selected.
```

Report Structure

This report can be used to understand:

- Who purchased the product
- Which order was placed
- Which product was purchased
- Which category the product belongs to
- Quantity purchased
- Amount generated

Sample Data Used

The outputs in this document are generated from the following sample records.

Category

Category_ID	Category_Name
1	Men Topwear
2	Women Ethnic Wear
3	Footwear
4	Accessories

Product

Product_ID	Product_Name	Brand_ID	Category_ID	Price
101	Roadster Cotton Shirt	1	1	1299
102	HRX Printed T-Shirt	2	1	699
103	Libas Anarkali Kurta	4	2	1899
104	Puma Running Shoes	3	3	3499
105	Puma Sports Cap	3	4	599

Customer

Customer_ID	First_Name	Last_Name
1001	Anitha	Kumar
1002	Rahul	Sharma
1003	Priya	Nair
1004	Karthik	Raja

Orders

Order_ID	Customer_ID	Order_Date	Order_Status	Total_Amount
5001	1001	05-OCT-26	Delivered	4496
5002	1002	12-OCT-26	Delivered	3499
5003	1003	18-OCT-26	Shipped	5895
5004	1001	24-OCT-26	Delivered	3297
5005	1004	30-OCT-26	Pending	1898

Order_Item

OrderItem_ID	Order_ID	Product_ID	Quantity	Unit_Price
1	5001	101	3	1299
2	5001	105	1	599
3	5002	104	1	3499
4	5003	103	2	1899
5	5003	102	3	699
6	5004	102	2	699
7	5004	103	1	1899
8	5005	101	1	1299
9	5005	105	1	599